

The Hong Kong Metropolitan University

School of Science and Technology

BSc (Hons) in Data Science and Artificial Intelligence

COMP S461F Data Science Project

2025-2026

**Improved Zero-Shot Classification through Efficient
CLIP-Based Vision-Language Fusion**

Group Leader:

YIP Ka Yiu 13720495

Group Members:

KONG Hau Ying 13143638

KWAN Yuen Man 13713897

LAM Ching Yiu 13793309

WU Yat Keung 13642323

Supervisor:

Dr. ZHANG Tiantian Daisy

24 April 2026

Acknowledgements

We thank Dr. ZHANG Tiantian Daisy for her guidance

Abstract

This research introduces a method for zero-shot image classification algorithm based on the CLIP model which balances high performance with computational efficiency. To address the high resource demands of standard models, we modified the image encoder to reduce latency and memory consumption. By optimizing the encoder's architecture, we achieve higher classification accuracy while significantly lowering GFLOPs. Experimental results across general, industrial and medical domains demonstrate that our model maintains better generalization and accuracy without using traditional large-scale encoders, making it ideal for resource constrained environments. Our evaluation scripts and performance logs are publicly hosted at <https://github.com/KaitlynYip/CompS461F-Group4>.

Contents

1 Abstract	3
2 Introduction	6
2.1 Research Objectives	7
3 Literature Review	8
3.1 Image Classification	8
3.2 Zero-shot Learning	8
3.3 CLIP-based Algorithm.....	9
4 Methodology	10
4.1 Implementation and Analysis	10
4.1.1 Backbone Selection	10
4.2 Challenges of Zero-Shot Generalization in Specialized Domains	10
4.2.1 Low-Rank Adaptation (LoRA) Strategy for Industrial Domain	10
4.2.2 Unfrozen Backbone and PubMedBERT for Medical Domain.....	11
4.3 Model Compression: Vit-S/16	11
4.4 Visual Feature Extraction with DINOv2.....	11
4.4.1 DINOv2 as the Visual Backbone.....	11
4.4.2 Feature Fusion Strategy Design	12
5 Experimental Data	13
5.1 Dataset Description	13
5.1.1 General Domain	13
5.1.2 Industrial Domain.....	16
5.1.3 Medical Domain	17
6 Experiment Results	19
6.1 Metric.....	19
6.2 General	21
6.3 Industrial	22
6.4 Medical	26
6.4.1 CheXray-Specific Evaluation Results	27
6.4.2 CheXray-Specific Enhanced Strategy and Results	28
7 Conclusion	29
7.1 Summary	29
7.1.1 Key enhancements	29
7.2 Limitations and future directions.....	29
7.2.1 Limitation	29
7.2.2 Future Directions	29

8	Appendix	30
8.1	Demo	30
8.1.1	General	30
8.1.2	Industrial	31
8.1.3	Medical.....	31
9	References	33

Introduction

Image classification is a fundamental task in computer vision that categorizes images into pre-defined classes by automatically analyzing and labeling them based on learned features such as shapes, textures, and patterns. While traditional machine learning methods were previously used, Convolutional Neural Networks (CNNs) have enabled better accurate and efficient feature extraction by pulling out key patterns from raw pixels, especially when trained on large datasets. [1] However, image classification need extensive labeled datasets and often struggles to classify unseen classes [2, 3], which limit their adaptability. Therefore, zero-shot classification overcomes these challenges by handling unseen categories without requiring many labeled examples. [4, 5]. This is crucial for applications like recognizing rare objects which can help us in everyday life. In medical field, it can analyze medical images like X-rays and CT scans to improve diagnostic accuracy by detecting disease in images that human observers might miss. This is not only improving the effectiveness of treatment, but also reducing the workload of doctors. Similarly in industrial field, it enables quality control by identifying defects in product to ensure the safety before reaching the customers. By addressing the large labeled datasets and enabling unseen class recognition, zero-shot classification enhances the scalability, practicality, and cost-effectiveness of image classification in real world applications.

Zero-shot learning (ZSL) allows a model to recognize new categories it has never seen before. It does this by using text information to help the model understand what a new image might look like [2]. Early methods used simple word embeddings or manual descriptions to connect images and text [6, 7], but these often struggled to work well in different situations. Modern methods now use generative techniques to make the model more fair, ensuring it doesn't just favor the categories it already knows [8, 9]. However, the specific fields like medical imaging and industrial inspection are still difficult. These areas require the model to see very small differences between objects, which is hard when the model is biased toward its original training data [9, 10].

Therefore, multimodal models such as CLIP has transformed the ZSL framework towards web-scale pre-training and prompt-based inference [11, 12]. While modern methods like WinCLIP and SPT-SEG have improved certain tasks such as anomaly detection [12, 13], the high computational cost (GFLOPs) are still too expensive with the limited resources. Even though there have been big improvements in performance [14, 15]. The large number of parameters means that we still need to focus on finding efficient, flexible solutions for real-time applications.

2.1 Research Objectives

The objective of this research is to develop a lightweight zero-shot image classification algorithm that achieves high performance across various domains. By modifying the image encoder, this study aims to enhance the classification accuracy, as well as reducing computational cost.

The Research Objectives are shown as follows:

- Propose light-weight zero shot image classification algorithm.
- Modify the image encoder to archive better performance.
- The experiments will be performed on the dataset from multiple domains

Literature Review

3.1 Image Classification

Convolutional Neural Networks (CNNs) started out from grid-processing models to specialized architectures designed to overcome specific classification and deployment constraints [3]. The focus of early developments was to make them more robust through various practices, such as regularization techniques like DisturbLabel [16], resolving overlapping semantic labels through a combination of CNN and RNN based architectures [17], and using Mask RCNNs to extend their ability to perform instance segmentation [18]. To overcome the limitations of early CNNs, later research focused on optimizing their depth. ResNet introduced the ability to successfully train deep networks without losing prediction accuracy [19] and Hybrid Dilated CNNs ensured that each pixel within each image contributed to the output result [20].

The focus of training networks has shifted significantly with the Vision Transformer (ViT), which utilizes image sequences like text to produce traditional models on large datasets [5]. For specialized domains like biological imaging, models such as InceptionV3 and DenseNet were developed to detect fine differentiated features [21]. Beyond structural changes, today's research emphasizes efficiency through transfer learning from large-scale datasets [22] and data augmentation [23]. Lightweight designs such as TurkNet [24] and Knowledge Distillation [25] have been created to enable high accuracy with limited resources. These strategies are often combined with Squeeze-and-Excitation mechanisms to dynamically recalibrate feature responses to optimize model efficiency.

3.2 Zero-shot Learning

ZSL allows models to identify unseen categories by transferring knowledge from related categories learned during training, such as attributes or text [2]. Initial studies founded the field through attribute based classification [6] and label embedding techniques can correlate image features with semantic labels [7]. To improve robustness, later works introduced intermediate embedding spaces [7, 26], semi-supervised frameworks [27], and generative models to create synthetic examples of unseen categories, which helps the model avoid over focusing on the classes it saw during training. [8, 9].

Recently, ZSL has expanded into specialized domains, such as medical and industrial images [4, 13]. However, there are a lot of challenges in these high stakes environments, including poor fine-grained discrimination and high computational costs [9, 28, 29].

Although models like CLIP work well for general tasks, they often struggle with specialized fields like medical or industrial [11, 10, 30]. This is because of the domain-specific data distributions and strict safety requirements, causing that more adaptation strategies are needed to use to ensure its reliability.

3.3 CLIP-based Algorithm

CLIP is a model developed by OpenAI that learns visual concepts from natural language supervision using large-scale image-text pairs from the internet. It uses image and text features through contrastive learning, enabling zero-shot classification and other vision tasks.

In the general domain, many studies have found ways to improve the overall performance of the CLIP model. Dong et al. [31] proved that fine-tuning with optimized hyperparameters increases accuracy, while RegionClip [32] uses pseudo-labeled pairs for better open-vocabulary object detection. Gaussian Discriminant Analysis has also been applied to improve few-shot and imbalanced learning [33] to increase the efficiency of CLIP without requiring full retraining.

For industrial applications, many companies are adapting CLIP to find anomalies. WinCLIP [12] combines prompts with images and segmentation to provide highly detailed analysis, while ZegCLIP [34] replace resource-intensive two-stage methods with a lightweight decoder. Further improvements in feature alignment include CLIP-AD's [35] dual-path architecture and PA-CLIP's [36] pseudo-anomaly framework for noise reduction. Moreover, Lu [37] optimized inference by pre-computing text embeddings, allowing efficient classification using only the image encoder.

In the medical domain, CLIP faces many challenges to overcome such as data scarcity and variable distribution. PubMedCLIP [38] shows an early effort to fine-tune on medical domain for improving VQA. Building on this, Zhao et al. [39] optimized medical CLIP for cross-modal tasks, resolving the alignment issues and providing a foundation for future medical vision-language correlations.

Methodology

4.1 Implementation and Analysis

4.1.1 Backbone Selection

Image encoder

We chose ViT-B/16 as our baseline image encoder due to its excellent transfer learning capabilities. Other common convolutional architectures, such as ConvNeXt-Base, are also powerful. However, they often require massive amounts of labeled data. In contrast, ViT-B/16 is more efficient and aligns better with vision-language models like CLIP, making it the ideal choice for applying in different domains.

Text encoder

Freezing the text encoder is important to preserve the pre-trained weights while only training the image encoder. For general and industrial domain, BERT-Base is used as it has been trained on various collection of text. However, in medical domain, PubMedBERT is preferred as it is pre-trained from scratch on biomedical literature which has a specialized understanding of clinical language that general models lack.

4.2 Challenges of Zero-Shot Generalization in Specialized Domains

Our baseline evaluation revealed a significant performance gap between general and specialized domains. While the initial model worked well on natural images, accuracy was low for industrial and medical tasks. This "domain gap" exists because general-purpose models lack the specific visual features required to identify detailed medical pathology or industrial anomalies. To bridge this divide, we implemented the following strategies.

4.2.1 Low-Rank Adaptation (LoRA) Strategy for Industrial Domain

In the industrial domain, the application of LoRA brought strong results. By freezing the vision backbone and only updating the adapter layers, we prevented the model from "memorizing" the background of training images. This allowed the model to focus specifically on relevant industrial defects.

As a result, the industrial performance was boosted by approximately 20%. This reveals that for distinct physical objects, parameter-efficient tuning is sufficient to cooperate with the DINOv2 and BERT encoders in the industrial domain.

4.2.2 Unfrozen Backbone and PubMedBERT for Medical Domain

The medical domain presented a much harder challenge, where medical X-rays contain subtle features that are difficult to distinguish. This is because the DINOv2 image encoder was pre-trained on general RGB images rather than grayscale medical imagery. Consequently, the standard LoRA approach proved less effective in this specialized field.

To address this, we implemented another strategy:

- **Text Encoder:** We switched the text encoder to **PubMedBERT** to provide specific medical terminology.
- **Image Encoder:** We unfroze the vision backbone to allow its weights to update alongside the LoRA layers.

By unfreezing all model parameters, the system was able to learn some more complex subtle features. This optimization resulted in an effective performance boost. It achieved a 40% gain in accuracy compared to the initial baseline. This demonstrates that for some specialized fields like the medical field, both specialized text encoders and backbone updates are essential.

4.3 Model Compression: ViT-S/16

The goal of this step is to improve computational efficiency by downsizing the baseline ViT-B/16 (86 million parameters) to the ViT-Small (ViT-S) model (22 million parameters). The ViT-S architecture uses 12 layers, 384 hidden dimensions, and 6 attention heads. This compression reduces the model's size and creates a trade-off in reducing the classification accuracy, precision, and recall as the efficiency improves. This experiment allows us to measure these performance differences and develop recovery strategies in the next step. [40]

4.4 Visual Feature Extraction with DINOv2

4.4.1 DINOv2 as the Visual Backbone

DINO family models are self-supervised vision foundation models developed by Meta AI that learn rich visual representations through self-distillation mechanisms without labeled data. The model employs a teacher-student architecture where the teacher network is updated from the student network via exponential moving average, with both processing different augmented views of the same image as input. This training paradigm enables the model to capture semantically rich features, particularly demonstrating precise localization capabilities for object boundaries in attention maps. [41]

DINOv2 is highly scalable, available in sizes from Small to Giant [42]. We incorporate its fine-grained features into the CLIP architecture to boost the visual-semantic alignment and zero-shot discriminability. Lighter versions such as ViT-S allow for high classification performance with tight computational budgets. The integration of DINO's visual representations

into the CLIP architecture can improve the quality of visual-semantic alignment, improve the feature discriminability and improve the zero-shot transfer capability to improve the zero-shot classification under limited computing resources.

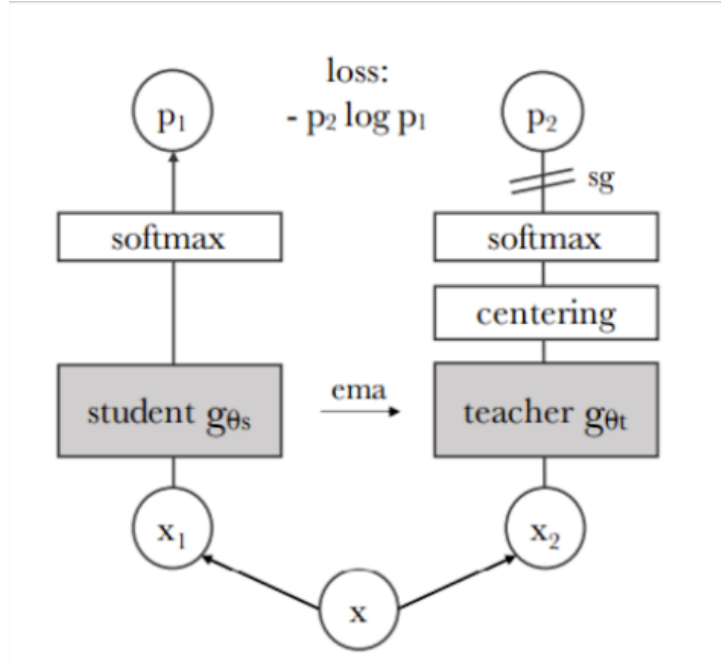


Figure 4.1: Framework of Dino
[41]

4.4.2 Feature Fusion Strategy Design

To recover performance lost from compression, this work replaces the standard CLIP image encoder with self-supervised DINO encoders. We apply and compare several feature fusion strategies including weighted fusion, attention mechanisms, and multi-scale aggregation based on DINO rich and unlabeled structural representations. The goal is to identify an optimal combination that enables lightweight models to match or exceed the zero-shot performance of the original ViT-B/16 baseline.

Experimental Data

5.1 Dataset Description

The following section provides an introduction to the dataset, elaborating on its organizational structure, core attributes, and essential characteristics. These are important for spotting patterns, creating effective models, and ensuring strong performance in image classification tasks, especially for complex models like CLIP.

The dataset covers three distinct domains: the General Domain, the Industrial Domain, and the Medical Domain which details are shown on the following pages.

5.1.1 General Domain

Caltech-101

It[43] contains 9,144 images across 101 diverse object categories. Each class has an average of about 90 images. All images are cropped and resized to 300x200 pixels. Some images are sourced from real-world photographs. Its small size makes it good for academic experiments.

Category	Train Images	Test Images
Object Categories		
Faces	314	121
Motorbikes	567	233
Airplanes	523	203
Cameras	268	106
Bicycles	292	120
Cars	568	218
Leopards	181	76
Chairs	298	105
Watches	170	65
Birds	292	119
Summary		
Total (101 categories)	6,738	2,406

Table 5.1: Caltech-101 Dataset Split (9,144 images across 101 categories)

CIFAR-10

It [44] consists of 60,000 low-resolution color images with 32x32 resized pixels across 10 classes of objects with 6,000 images per class. It is developed by the Canadian Institute, CIFAR-10 remains common for prototyping CNN and remains available for research.

Category	Train Images	Test Images
Object Categories		
Airplane	5,000	1,000
Automobile	5,000	1,000
Bird	5,000	1,000
Cat	5,000	1,000
Deer	5,000	1,000
Dog	5,000	1,000
Frog	5,000	1,000
Horse	5,000	1,000
Ship	5,000	1,000
Truck	5,000	1,000
Summary		
Total	50,000	10,000

Table 5.2: Distribution of CIFAR-10 Dataset (60,000 images across 10 categories)



Figure 5.1: Example images from the CIFAR-10 and Caltech-101 Dataset

PASCAL VOC 2012

It [45] involves of 20 object categories and total of 17,125 images. This dataset is created to assess algorithms for tasks including object identification, detection, segmentation, and similar activities.

Table 5.3: Distribution of PASCAL VOC 2012 (17,125 images across 20 categories)

Category	Train	Test
Aeroplane	546	137
Bicycle	359	90
Bird	614	153
Boat	369	93
Bottle	313	78
Bus	324	81
Car	838	157
Cat	838	109
Chair	438	109
Cow	261	65
Diningtable	230	57
Dog	963	241
Horse	372	93
Motorbike	345	86
Person	5,512	1,378
Pottedplant	222	56
Sheep	275	69
Sofa	314	78
Train	426	107
Tvmonitor	351	88

Total: 13,700 training images and 3,425 testing images.



(a) Cat



(b) Aeroplane

Figure 5.2: Example images from PASCAL VOC 2012

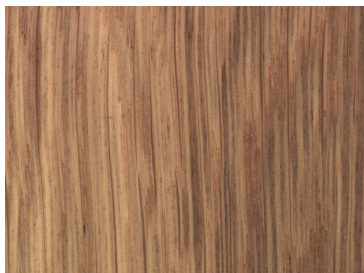
5.1.2 Industrial Domain

MVTec AD

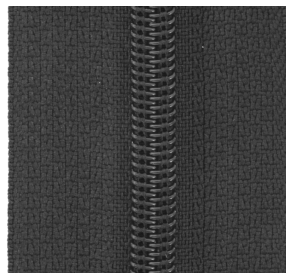
It contains 5,354 high-resolution color images across 15 categories, including 10 object classes (e.g., bottle, capsule, metal nut) and 5 texture classes (e.g., carpet, leather, wood). All images were captured under controlled industrial conditions with resolutions between 700×700 and 1024×1024 pixels, closely simulating industrial visual inspection scenarios where only a limited number of defective samples are available[46, 28].

Table 5.4: Statistical overview of the MVTec AD dataset [28]. The dataset includes 15 categories (5 textures and 10 objects) with a total of 5,354 high-resolution images.

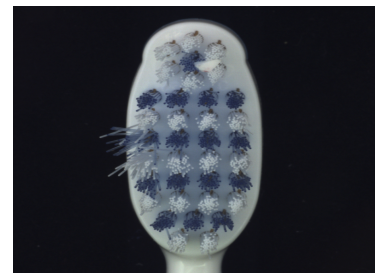
Category	Train Images	Test (Good)	Test (Defective)	Image Size (px)
Textures				
Carpet	280	28	89	1024×1024
Grid	264	21	57	1024×1024
Leather	245	32	92	1024×1024
Tile	230	33	84	840×840
Wood	247	19	60	1024×1024
Objects				
Bottle	209	20	63	900×900
Cable	224	58	92	1024×1024
Capsule	219	23	109	1000×1000
Hazelnut	391	40	70	1024×1024
Metal Nut	220	22	93	700×700
Pill	267	26	141	800×800
Screw	320	41	119	1024×1024
Toothbrush	60	12	30	1024×1024
Transistor	213	60	40	1024×1024
Zipper	240	32	119	1024×1024
Total	3,629	467	1,258	—



(a) Wood



(b) Zipper



(c) Toothbrush

Figure 5.3: Example images from MVTec AD

5.1.3 Medical Domain

Covid-chestxray-Dataset

This open-source collection [47] primarily focuses on COVID-19, which accounts for approximately 80% of the images. To improve training efficiency, we consolidated various bacterial and viral pneumonia subtypes into two primary categories. This allows the model to focus on identifying major pneumonia types rather than minor class differences.

Classes	Number of data
No Finding	22
Pneumonia	81
Pneumonia/Aspiration	1
Pneumonia/Bacterial	4
Pneumonia/Bacterial/Chlamydomphila	3
Pneumonia/Bacterial/E.Coli	4
Pneumonia/Bacterial/Klebsiella	10
Pneumonia/Bacterial/Legionella	10
Pneumonia/Bacterial/Mycoplasma	11
Pneumonia/Bacterial/Nocardia	8
Pneumonia/Bacterial/Staphylococcus/MRSA	1
Pneumonia/Bacterial/Streptococcus	22
Pneumonia/Fungal/Aspergillosis	2
Pneumonia/Fungal/Pneumocystis	30
Pneumonia/Lipoid	13
Pneumonia/Viral/COVID-19	584
Pneumonia/Viral/Herpes	3
Pneumonia/Viral/Influenza	5
Pneumonia/Viral/Influenza/H1N1	2
Pneumonia/Viral/MERS-CoV	10
Pneumonia/Viral/SARS	16
Pneumonia/Viral/Varicella	6
Tuberculosis	83
todo	18
Unknown	1

Table 5.5: Distribution of covid-chestxray-dataset from the given csv file in the official github repository

Dataset splitting and OOD Formulation

The training and validation sets consist of four well-defined categories: Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia (excluding COVID-19). We merged the respiratory pneumonia subtypes because they share similar radiographic patterns and clinical management.

Remaining pneumonia types are excluded from training and treated as out-of-distribution (OOD) test classes. This open-set approach reflects realistic clinical scenarios where models encounter rare or unseen diseases, following established protocols in recent domain generalization studies [48].

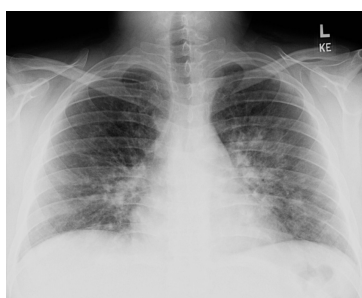
Class distribution after data splitting:

(a) Training / Validation Set		(b) Out-of-Distribution (OOD) Test Set	
Class	Count	Class	Count
COVID-19	563	Generic / Ambiguous Pneumonia	81
Bacterial	73	Pneumocystis	30
Viral (Except COVID-19)	42	Tuberculosis	18
Normal	22	Lipoid Pneumonia	13
		Aspergillosis	2
		Aspiration Pneumonia	1

Table 5.6: Class distribution of the training/validation and OOD test sets.



(a) frontal view(unknown)



(b) frontal view(Pneumonia)



(c) lateral view(Pneumonia)

Figure 5.4: Example images from covid-chestxray-dataset

Experiment Results

6.1 Metric

It is critical to evaluate the performance of various zero-shot CLIP-based vision-language fusion models. Therefore, key multi-class image classification metrics should be calculated to evaluate the effectiveness of the model.

Confusion Matrix

For multi-class classification, the confusion matrix is extended to an $n \times n$ matrix (where n is the number of classes), with true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) aggregated or computed per class.

Accuracy

Accuracy measures the overall percentage of correct predictions made by the model across all classes which helps evaluate how well the vision-language fusion captures for the unseen classes.

$$\text{Accuracy} = \frac{\sum_{i=1}^k TP_i}{N} = \frac{\text{Correct Predictions}}{\text{All Predictions}}$$

where i = Index of the class category

Precision

Precision evaluates the accuracy of positive classifications by calculating the percentage of positive predictions that are actually true which can determine the reliability which helps to highlight how well the CLIP model avoids misclassifying non-target images as others class.

$$\text{Precision}_i = \frac{TP}{TP + FP}$$

Recall(Sensitivity)

Recall, as known as the sensitivity calculates the percentage of real positives that the model correctly identified. It is important for ensuring underrepresented classes are not missed.

$$\text{Recall}_i = \frac{TP}{TP + FN}$$

Area Under the Receiver Operating Characteristic Curve (AUROC)

AUROC measures how well the model distinguishes between different classes. In OOD detection, it represents the probability that a randomly selected OOD image will have a higher anomaly (energy) score than a normal image. A score of 1.0 means perfect detection, and 0.5 indicates the model is just guessing.

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t)) dt$$

6.2 General

Caltech-101 and CIFAR-10

Through using prompt engineering and text templates, we generated robust embeddings for the CIFAR-10 and Caltech-101 datasets. DINOv2-Small and BERT are used for vision and text respectively. The model achieved impressive zero-shot results. This demonstrates the model’s strong ability to generalise to unseen labels without additional training. These results prove that our architecture is highly effective for general domain data, establishing a clear baseline for our research. In practical terms, this means the model can recognise and relate visual concepts even when it has never explicitly seen those categories during training. Such capability highlights the strength of combining high-quality visual representations with well-structured textual prompts and allowing the system to bridge the gap between images and language in a flexible and scalable way.

Image Encoder	Accuracy	Total GFLOPs	Throughput (FPS)	Latency (GPU)	Trainable %
Vit-base	96.48%	35.2	71.16 FPS	14.05 ms	100.00%
Vit-Small	94.38%	4.24	178.72 FPS	5.49 ms	100.00%
Dino V2 Small	96.54%	4.60	110.99 FPS	9.01 ms	100.00%

Table 6.1: Classification performance on the Caltech-101

Image Encoder	Accuracy	Total GFLOPs	Throughput (FPS)	Latency (GPU)	Trainable %
Vit-base	96.48%	16.87	57.07 FPS	17.52 ms	100.00%
Vit-Small	97.02%	4.25	95.77 FPS	10.44 ms	100.00%
Dino V2 Small	97.34%	5.53	61.96 FPS	16.14 ms	100.00%

Table 6.2: Classification performance on the CIFAR-10

PASCAL VOC 2012

However, the results from this dataset are worse because of imbalance in the data set. The distribution is heavily skewed toward the "person" category, which comprises 6,890 images, where the minority "pottedplant" class contains only 278 images. This keeps the model from being able to generalize well to all categories because the model is not exposed to enough images of each class. As a result, the model predicts the majority class when it has to use its predictions on unseen or new data.

Consequently, the model struggles to learn meaningful features for underrepresented classes and it leading to poorer prediction accuracy and reduced reliability when identifying those minority categories in unseen data.

Image Encoder	Accuracy	Total GFLOPs	Throughput (FPS)	Latency (GPU)	Trainable %
Vit-base	80.29%	16.87	72.58 FPS	13.78 ms	100.00%
Vit-Small	80.58%	4.25	182.04 FPS	5.49 ms	100.00%
Dino V2 Small	75.80%	5.53	162.08 FPS	6.17 ms	100.00%

Table 6.3: Classification performance on the PASCAL VOC 2012

6.3 Industrial

MVTEC Anomaly Detection Dataset

By using the test set for evaluation, the results of confusion matrix analysis in our experiments show that the ViT-B-16 baseline achieved 68.99% accuracy in Zero-Shot classification. The model showed good performance on simple categories such as pill, screw, and zipper, which achieved a near 100% precision. However, some categories, such as capsule and cable show a poor result. This is likely because the CLIP did not train on the industrial aspect before.

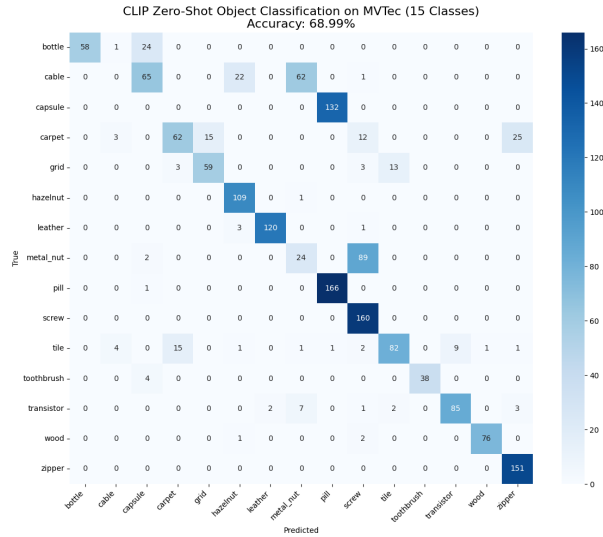


Figure 6.1: Confusion Matrix for Zero-Shot CLIP on MVTEC AD(Baseline Accuracy: 68.99%)

Therefore, we applied LoRA for efficient fine-tuning on the three image encoder. The comparison of setup and accuracy is shown below:

Image Encoder	Accuracy	Total GFLOPs	Throughput	Latency (GPU)	Trainable %
ViT-base	97.16%	16.98	62.66 FPS	15.96 ms	1.36%
ViT-small	92.93%	4.30	115.92 FPS	8.63 ms	2.65%
Dino V2 Small	95.71%	12.58	82.65 FPS	82.65 ms	2.61%

Table 6.4: Performance Comparison of Image Encoders

For the implementation of LoRA in the ViT-B, it shows a significant improvement performance

compared to the zero-shot baseline. The adapted model achieves an overall classification accuracy of 97.16% across the 1,725 test images in the MVTec AD dataset.

Category	Precision	Recall	F1-score	Support
bottle	1.0000	0.9880	0.9939	83
cable	1.0000	0.9933	0.9967	150
capsule	1.0000	1.0000	1.0000	132
carpet	1.0000	1.0000	1.0000	117
grid	1.0000	1.0000	1.0000	78
hazelnut	1.0000	1.0000	1.0000	110
leather	1.0000	1.0000	1.0000	124
metal_nut	0.9914	1.0000	0.9957	115
pill	1.0000	0.7305	0.8443	167
screw	1.0000	1.0000	1.0000	160
tile	1.0000	0.9829	0.9914	117
toothbrush	0.4773	1.0000	0.6462	42
transistor	0.9804	1.0000	0.9901	100
wood	1.0000	1.0000	1.0000	79
zipper	1.0000	1.0000	1.0000	151
Accuracy			0.9716	1725
Macro Avg	0.9633	0.9796	0.9639	1725
Weighted Avg	0.9856	0.9716	0.9743	1725

Table 6.5: Classification Performance per Category (Vit-B/16)

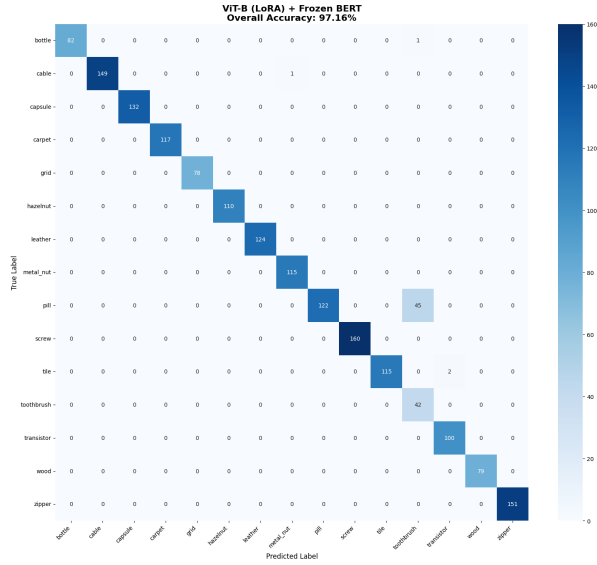


Figure 6.2: Confusion Matrix of ViTB model(Accuracy: 97.16%)

To optimize the trade off between accuracy and computational cost, the ViT-S was applied. As shown in Table 6.6, there is a reduction in classification accuracy compared to the ViT-B baseline. This performance decline is directly attributed to the overall loss of optimized efficiency in computation, by reducing the total GFLOPs from 16.9834 to 4.3087.

Category	Precision	Recall	F1-score	Support
bottle	1.0000	1.0000	1.0000	83
cable	1.0000	0.9933	0.9967	150
capsule	1.0000	0.9848	0.9924	132
carpet	1.0000	0.9829	0.9914	117
grid	1.0000	1.0000	1.0000	78
hazelnut	1.0000	1.0000	1.0000	110
leather	1.0000	1.0000	1.0000	124
metal_nut	0.9914	1.0000	0.9957	115
pill	1.0000	0.3293	0.4955	167
screw	1.0000	1.0000	1.0000	160
tile	1.0000	0.9573	0.9782	117
toothbrush	0.2658	1.0000	0.4200	42
transistor	0.9524	1.0000	0.9756	100
wood	1.0000	1.0000	1.0000	79
zipper	1.0000	1.0000	1.0000	151
Accuracy			0.9293	1725
Macro Avg	0.9473	0.9498	0.9230	1725
Weighted Avg	0.9788	0.9293	0.9324	1725

Table 6.6: Classification Performance per Category (Vit-S/16)

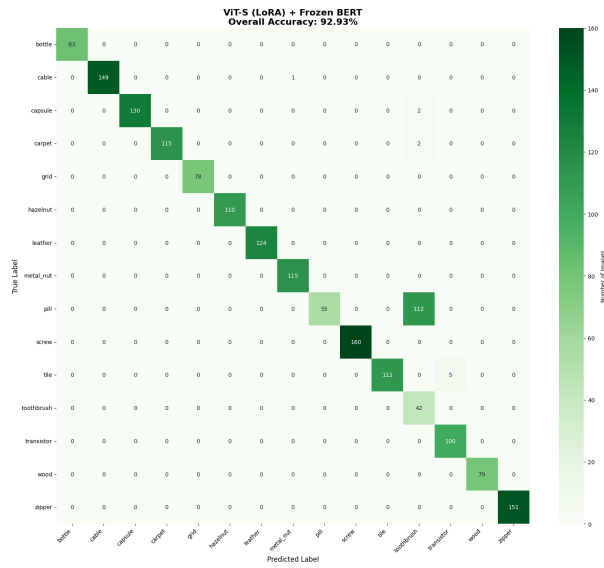


Figure 6.3: Confusion Matrix of ViT-S model(Accuracy: 92.93%)

Finally, the DINOv2 architecture was implemented to achieve an optimal balance between computational efficiency and predictive performance. As summarized in the Table 6.4, DINOv2 achieves an accuracy of 95.71% which is better than ViT-S. Regarding to the computational cost, DINOv2 requires 12.5800 GFLOPs, showing a large reduction from the ViT-B (16.9834 GFLOPs) while providing an improvement in accuracy over ViT-S. This shows that DINOv2 is the most efficient model, successfully bridging the performance gap between the lightweight ViT-S and the more computationally demanding ViT-B.

Category	Precision	Recall	F1-score	Support
bottle	1.0000	1.0000	1.0000	83
cable	1.0000	0.9867	0.9933	150
capsule	1.0000	1.0000	1.0000	132
carpet	1.0000	1.0000	1.0000	117
grid	1.0000	1.0000	1.0000	78
hazelnut	1.0000	1.0000	1.0000	110
leather	1.0000	1.0000	1.0000	124
metal_nut	0.9829	1.0000	0.9914	115
pill	1.0000	0.5689	0.7252	167
screw	1.0000	1.0000	1.0000	160
tile	1.0000	1.0000	1.0000	117
toothbrush	0.3684	1.0000	0.5385	42
transistor	1.0000	1.0000	1.0000	100
wood	1.0000	1.0000	1.0000	79
zipper	1.0000	1.0000	1.0000	151
Accuracy			0.9571	1725
Macro Avg	0.9568	0.9704	0.9499	1725
Weighted Avg	0.9835	0.9571	0.9610	1725

Table 6.7: Classification Performance per Category (DinoV2-Small)

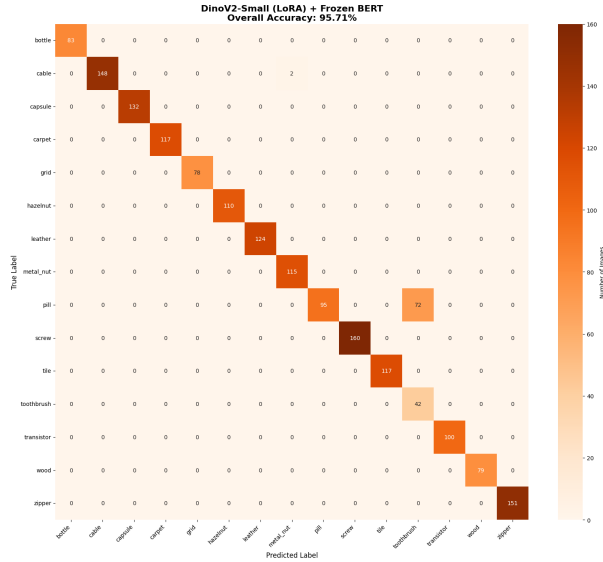


Figure 6.4: Confusion Matrix of DinoV2 model(Accuracy: 95.71%)

Prediction examples from the MVTec AD dataset for each class are shown in the 3x5 grid below. Each image was randomly selected from the test dataset. With the 'True' labels matching the 'Pred' labels, it can highlight the high reliability to distinguish industrial images between

15 categories.

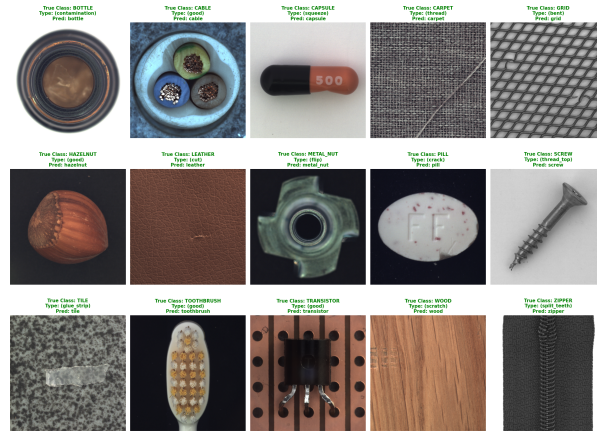


Figure 6.5: Image result on MVTec AD

6.4 Medical

Table 6.8 summarizes the classification performance on the medical validation set under zero-shot and LoRA-adapted settings. In the zero-shot configuration, CLIP exhibits limited performance, achieving an overall accuracy of 29.5%. While COVID-19 cases are detected with high precision, recall remains low, and performance on non-COVID classes is near zero. This indicates that, without adaptation, the pretrained model struggles to distinguish radiographic patterns associated with different pneumonia types.

After applying LoRA adaptation, the classification performance improves across all classes. Overall accuracy increases to 69.52%, with the improvements in across recall and F1-score for Viral (Except COVID-19) and Bacterial pneumonia.

Table 6.8: Classification performance on the medical validation set.

	Zero-shot CLIP ViT-B/16				CLIP ViT-B/16 + LoRA Adapter			
	Precision	Recall	F1	Support	Precision	Recall	F1	Support
Normal	0.00	0.00	0.00	2	0.08	0.50	0.14	2
Viral (Except COVID-19)	0.00	0.00	0.00	3	0.38	1	0.55	3
Bacterial	0.03	0.08	0.04	13	0.43	0.54	0.48	13
COVID-19	0.94	0.34	0.50	87	0.89	0.71	0.79	87
Accuracy	29.5%				69.52%			

6.4.1 CheXray-Specific Evaluation Results

Out Of Distribution Test

To see how well our models handle pneumonia cases they have not encountered before, we compared three different models using the AUROC metric. This metric is useful in monitoring medical tasks because it is more reliable than simple accuracy. AUROC measures how effectively a model tells the difference between healthy and diseased scans. This gives us a clear understanding of the model’s ability to identify pneumonia when it faces out-of-distribution (OOD) data that was not included in the original training set.

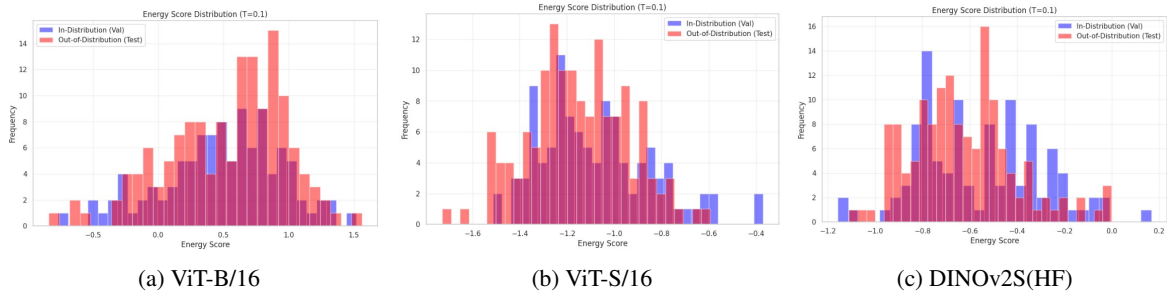


Figure 6.6

Image Encoder	Val Acc.	GFLOPs	Throughput	Latency	Trainable %	AUROC
ViT-Base	69.52%	16.87	69.78 FPS	14.33 ms	100.00%	0.47
ViT-Small	69.52%	4.25	101.08 FPS	9.89 ms	100.00%	0.57
DINOv2-Small (HF)	67.62%	5.53	105.17 FPS	9.51 ms	100.00%	0.58

Table 6.9: Performance Comparison of Vision Transformer Backbones

Although ViT-Small and ViT-Base had the highest validation accuracy (69.52%), DINOv2-Small (HF) generalized better, reaching the highest Test AUROC (0.58). With low latency (9.51 ms) and high throughput (105.17 FPS), DINOv2-Small is a more balanced and efficient choice. It is suited for medical tasks where reliability and speed are most important.

We found that using the DINOv2-Small encoder performed about the same as our baseline. Because this change offered little improvement, we chose to upgrade the overall model architecture instead.

6.4.2 CheXray-Specific Enhanced Strategy and Results

Balancing Frequency and Balancing Difficulty with Cost-Sensitive Learning

To increase the reliability of the proposed model, two approaches were implemented for dealing with the problem for the imbalanced dataset and difficult images. First of all, since 80% of images belong to the COVID-19 class. To prevent the model from assigning class based on frequency, weight was assigned to the less frequent classes. Thus, it is possible for the model to study characteristics rather than frequencies, which helps prevent the scenario when the artificial intelligence system predicts all cases belonging to the most common class.

Secondly, besides frequency, the approach considered cost sensitivity with high weight (13.0) assigned to the COVID-19 class. This approach created a penalty area during training, given the similarity of the texture of COVID-19 and Viral Pneumonia.

Optimizing LoRA Efficiency for Medical Generalization

The parameter r drives the amount of additional knowledge that can be learned by the adapter. In our initial setting, we used $r = 16$; however, this led to the fact that the network learned the specific features and noise of particular images, which posed the risk of overfitting, where the network memorized certain images rather than general features of medicine.

In order to enhance the performance of the network on out-of-set labels, we decreased r to 8. The decrease of the parameter’s value makes the network learn only the significant medical features since it limits its capacity. This serves as regularization for the network to better handle images of rare diseases or from other sources.

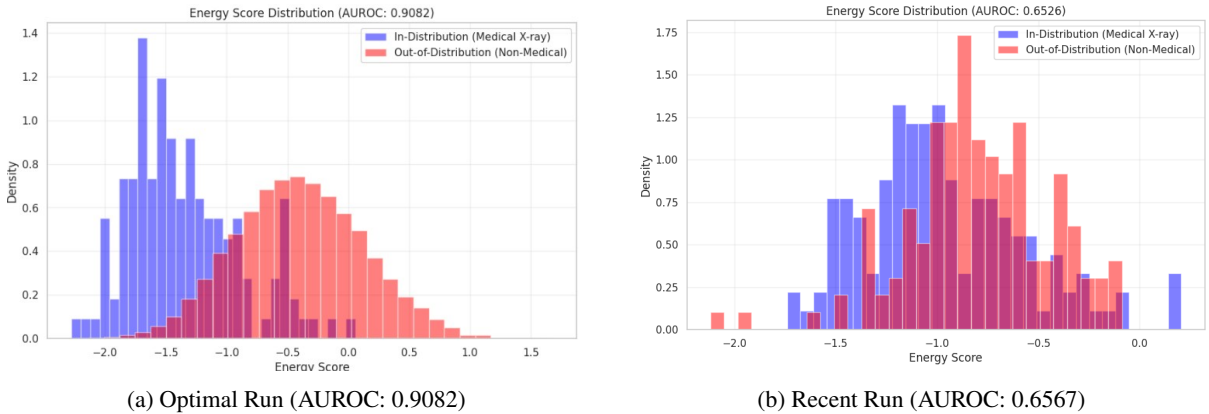


Figure 6.7: OOD detection performance comparison between the highest recorded metric and the most recent evaluation session using identical hyperparameter settings.

Enhanced Image Encoder	Val Acc.	GFLOPs	Throughput	Latency	Trainable %	AUROC
DINOv2-Small(HF)	67.85%	5.52	124.87 FPS	8.01 ms	100.00%	0.91

Table 6.10: Performance Comparison of Vision Transformer Backbones

Conclusion

7.1 Summary

We developed and improved three zero-shot classification models for General, Industrial, and Medical use. By using CLIP-based vision-language fusion, these models can identify objects or textures that they have never seen during training.

7.1.1 Key enhancements

The difficulty we encountered in the medical field was due to insufficient data and rare diseases. We resolved this problem by using cost-sensitive learning and LoRA tuning, giving more importance to the rare classes, and creating a penalty area for COVID-19 in order to enforce the model to capture medical textures.

In industry field, we demonstrated that efficient tuning through LoRA leads to good performance while saving time and money.

7.2 Limitations and future directions

7.2.1 Limitation

We are now working towards closing the Domain-Specific Semantic Gaps. Although DINOv2 effectively extracts structural aspects of an image well, zero-shot language comprehension will continue to rely on CLIP's text encoder. In medical domain, generic language models can cause semantic ambiguity, hindering the system's ability to distinguish visually similar.

7.2.2 Future Directions

Our objective is to develop this framework as a modular, plug-in system optimized for rapid deployment across different domains with minimal retraining. To demonstrate extensibility, we will also be testing the Bridge in new area such as Agriculture and Environmental Conservation.

Additionally, we are focusing on On-Device Optimization to democratize access to advanced AI. By porting the model from high-end GPUs to edge devices, we aim to enable local inference on mid-range smartphones. This approach enables the application to function offline, letting users to use it even without a steady internet connection.

Appendix

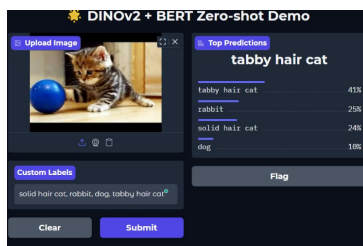
8.1 Demo

8.1.1 General

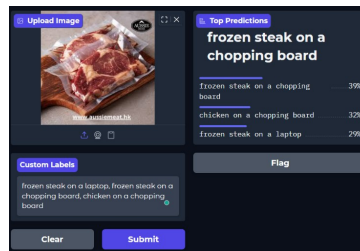
We used Caltech-101 and CIFAR-10 for this demo because both showed great results from our experiments. To overcome their individual trade-offs, we combined them because the CIFAR-10 provides enough data for general reliability, whereas Caltech-101 provides the complex details missing in low-resolution images. By merging these datasets, the model becomes more well-rounded and detail-oriented.

You may visit our demo model for general domain at the following link:

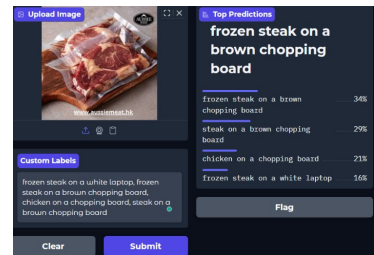
https://huggingface.co/spaces/Kaitlyn080544/COMPS461F_Group4_General_Demo



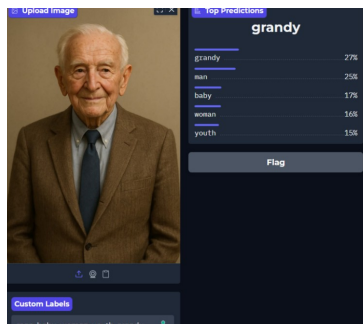
(a) Demo 1



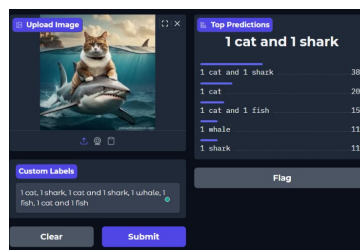
(b) Demo 2



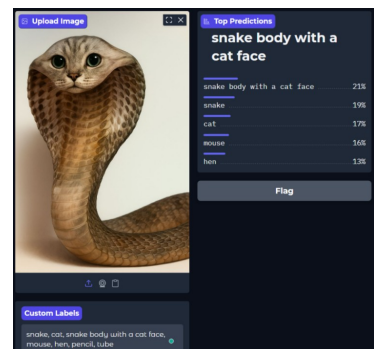
(c) Demo 3



(d) Demo 4



(e) Demo 5



(f) Demo 6

Figure 8.1: Zero-shot Classification Demos in the General Domain

8.1.2 Industrial

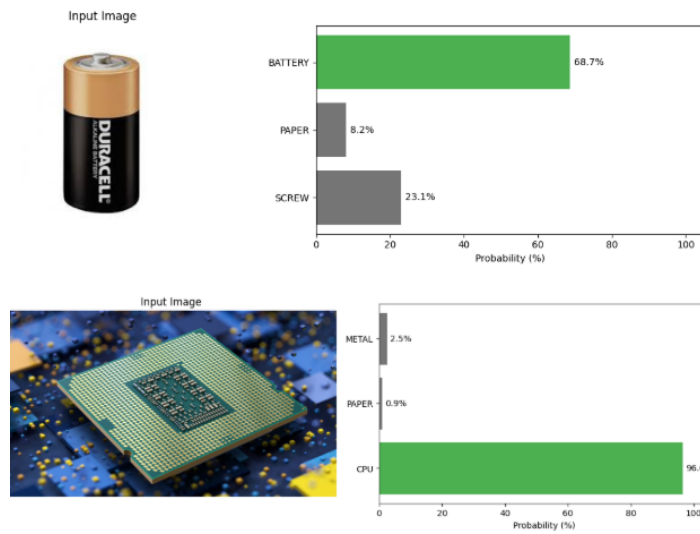


Figure 8.2: Zero-shot Demo in industrial domain

8.1.3 Medical

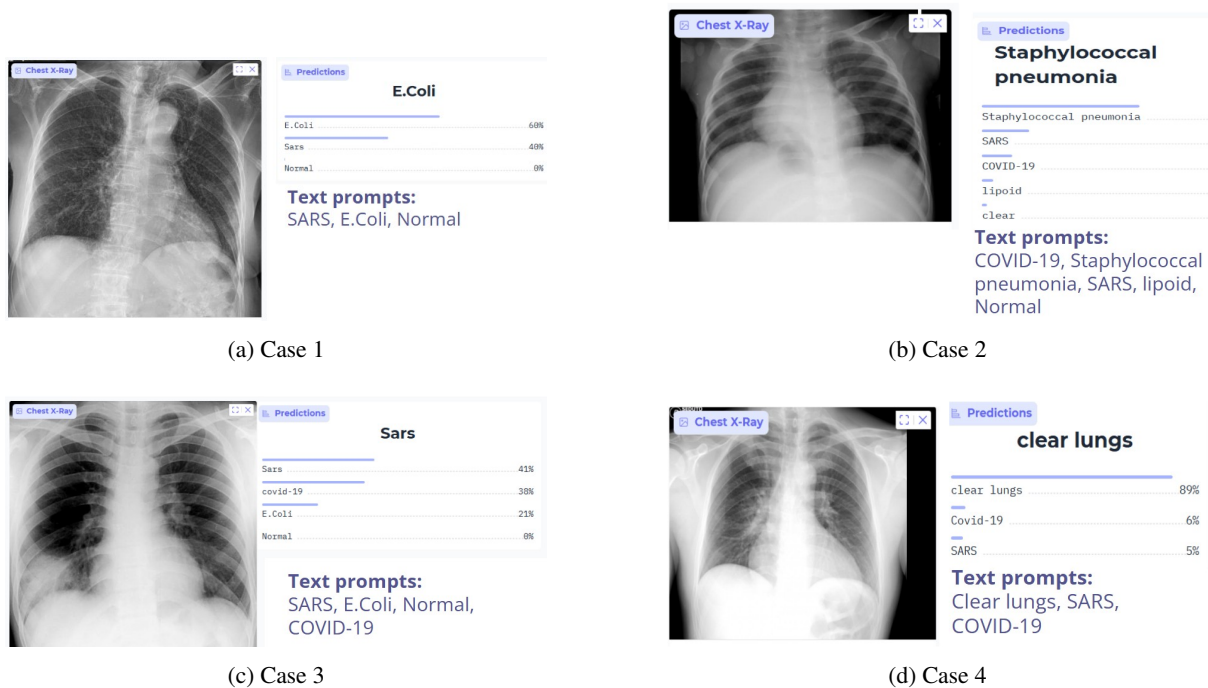
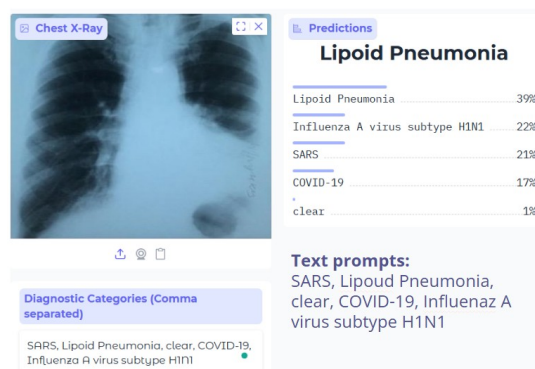


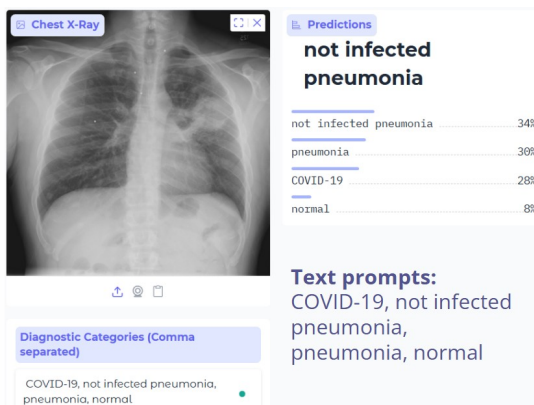
Figure 8.3: Medical Domain Zero-shot Classification Examples (Part 1)



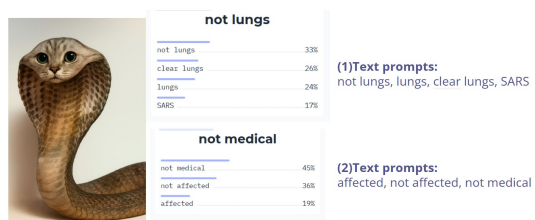
(e) Case 5



(f) Case 6



(g) Case 7



(h) Case 8

Figure 8.3: Medical Domain Zero-shot Classification Examples (Part 2)

References

- [1] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25, 2012.
- [2] Hee E. Kim, Alejandro Cosa-Linan, Nandhini Santhanam, Mahboubeh Jannesari, Mate E. Maros, and Thomas Ganslandt. Transfer learning for medical image classification: a literature review. *BMC Medical Imaging*, 22(1):69, 4 2022.
- [3] Waseem Rawat and Zenghui Wang. Deep convolutional neural networks for image classification: A comprehensive review. *Neural Computation*, 29(9):2352–2449, 2017.
- [4] Jiaxiang Liu, Tianxiang Hu, Yan Zhang, Xiaotang Gai, Yang Feng, and Zuozhu Liu. A chatgpt aided explainable framework for zero-shot medical image diagnosis, 2023.
- [5] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- [6] Christoph H Lampert, Hannes Nickisch, and Stefan Harmeling. Learning to detect unseen object classes by between-class attribute transfer. In *2009 IEEE conference on computer vision and pattern recognition*, pages 951–958. IEEE, 2009.
- [7] Zeynep Akata, Florent Perronnin, Zaid Harchaoui, and Cordelia Schmid. Label-embedding for attribute-based classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 819–826, 2013.
- [8] Yongqin Xian, Zeynep Akata, Gaurav Sharma, Quynh Nguyen, Matthias Hein, and Bernt Schiele. Latent embeddings for zero-shot classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 69–77, 2016.
- [9] Yongqin Xian, Tobias Lorenz, Bernt Schiele, and Zeynep Akata. Feature generating networks for zero-shot learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5542–5551, 2018.
- [10] Sedigheh Eslami, Gerard de Melo, and Christoph Meinel. Does clip benefit visual question answering in the medical domain as much as it does in the general domain?, 2021.
- [11] Jongseong Jang, Daeun Kyung, Seung Hwan Kim, Honglak Lee, Kyunghoon Bae, and Edward Choi. chatgptfically improving zero-shot x-ray pathology classification via fine-tuning pre-trained image-text encoders. *Scientific Reports*, 14(1), October 2024.
- [12] Jongheon Jeong, Yang Zou, Taewan Kim, Dongqing Zhang, Avinash Ravichandran, and Onkar Dabeer. Winclip: Zero-/few-shot anomaly classification and segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19606–19616, 2023.
- [13] Wenhao Xu, Rongtao Xu, Changwei Wang, Shibiao Xu, Li Guo, Man Zhang, and Xiaopeng Zhang. Spectral prompt tuning: Unveiling unseen classes for zero-shot semantic segmentation. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 6369–6377, 2024.

- [14] Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Robust fine-tuning of zero-shot models. *arXiv preprint arXiv:2109.01903*, 2022.
- [15] Xiaohua Zhai, Xiao Wang, Basil Mustafa, Andreas Steiner, Daniel Keysers, Alexander Kolesnikov, and Lucas Beyer. Clip itself is a strong fine-tuner: Achieving 85.7% and 88.0% top-1 accuracy with vit-b and vit-l on imagenet. *arXiv preprint arXiv:2212.06138*, 2022.
- [16] Lingxi Xie, Jingdong Wang, Zhen Wei, Meng Wang, and Qi Tian. Disturblabel: Regularizing cnn on the loss layer. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4753–4762, 2016.
- [17] Jiang Wang, Yi Yang, Junhua Mao, Zhiheng Huang, Chang Huang, and Wei Xu. Cnn-rnn: A unified framework for multi-label image classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2285–2294, 2016.
- [18] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969, 2017.
- [19] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [20] Xinyu Lei, Hongguang Pan, and Xiangdong Huang. A dilated cnn model for image classification. *IEEE access*, 7:124087–124095, 2019.
- [21] Wenyan Zhou, Hao Wang, and Zhibo Wan. Ore image classification based on improved cnn. *Computers and Electrical Engineering*, 99:107819, 2022.
- [22] Connor Shorten and Taghi M Khoshgohfar. A survey on image data augmentation for deep learning. *Journal of Big Data*, 6(1):60, 2019.
- [23] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7132–7141, 2018.
- [24] Mohammed Q Alkhatib, Mina Al-Saad, Nour Aburaed, Saeed Almansoori, Jaime Zabalza, Stephen Marshall, and Hussain Al-Ahmad. Tri-cnn: A three branch model for hyperspectral image classification. *Remote Sensing*, 15(2):316, 2023.
- [25] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- [26] Yanming Guo, Yu Liu, Ard Oerlemans, Songyang Lao, Song Wu, and Michael S. Lew. Deep learning for visual understanding: A review. *Neurocomputing*, 187:27–48, 2016. Recent Developments on Deep Big Vision.
- [27] Xin Li, Yuhong Guo, and Dale Schuurmans. Semi-supervised zero-shot classification with label representation learning. In *Proceedings of the IEEE international conference on computer vision*, pages 4211–4219, 2015.

- [28] Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger. Mvtec ad—a comprehensive real-world dataset for unsupervised anomaly detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9592–9600, 2019.
- [29] Edmon Begoli, Tanmoy Bhattacharya, and Dimitri Kusnezov. The need for uncertainty quantification in machine-assisted medical decision making. *Nature Machine Intelligence*, 1(1):20–23, January 2019. Publisher Copyright: © 2019, This is a U.S. government work and not under copyright protection in the U.S.; foreign copyright protection may apply.
- [30] Qing Li, Weidong Cai, Xiaogang Wang, Yun Zhou, David Dagan Feng, and Mei Chen. Medical image classification with convolutional neural network. In *2014 13th international conference on control automation robotics & vision (ICARCV)*, pages 844–848. IEEE, 2014.
- [31] Xiaoyi Dong, Jianmin Bao, Ting Zhang, Dongdong Chen, Shuyang Gu, Weiming Zhang, Lu Yuan, Dong Chen, Fang Wen, and Nenghai Yu. Clip itself is a strong fine-tuner: Achieving 85.7% and 88.0% top-1 accuracy with vit-b and vit-l on imagenet. *arXiv preprint arXiv:2212.06138*, December 2022.
- [32] Yiwu Zhong, Jianwei Yang, Pengchuan Zhang, Chunyuan Li, Noel Codella, Liunan Harold Li, Luwei Zhou, Xiyang Dai, Lu Yuan, Yin Li, and Jianfeng Gao. Regionclip: Region-based language-image pretraining. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16793–16803, 2022.
- [33] Zhengbo Wang, Jian Liang, Lijun Sheng, Ran He, Zilei Wang, and Tieniu Tan. A hard-to-beat baseline for training-free clip-based adaptation. In *International Conference on Learning Representations*, 2024.
- [34] Ziqin Zhou, Yinjie Lei, Bowen Zhang, Lingqiao Liu, and Yifan Liu. Zegclip: Towards adapting clip for zero-shot semantic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11175–11185, 2023.
- [35] Xuhai Chen, Jiangning Zhang, Guanzhong Tian, Haoyang He, Wuhao Zhang, Yabiao Wang, Chengjie Wang, and Yong Liu. Clip-ad: A language-guided staged dual-path model for zero-shot anomaly detection. In *International Joint Conference on Artificial Intelligence*, pages 17–33. Springer, 2024.
- [36] Yurui Pan, Lidong Wang, Yuchao Chen, Wenbing Zhu, Bo Peng, and Mingmin Chi. Pa-clip: Enhancing zero-shot anomaly detection through pseudo-anomaly awareness. *arXiv preprint arXiv:2503.01292*, 2025.
- [37] Jin Lu. Few-shot learning on edge devices using clip: A resource-efficient approach for image classification. *Information Technology and Control*, 53(3):833–845, 2024.
- [38] Shima Eslami, Fatemeh Asadollahi, and Ali Etemad. Are we making use of the best feature extraction techniques? In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pages 3775–3784, 2021.
- [39] Zihao Zhao, Yuxiao Liu, Han Wu, Mei Wang, Yonghao Li, Sheng Wang, Lin Teng, Disheng Liu, Zhiming Cui, Qian Wang, and Dinggang Shen. Clip in medical imaging: A survey. *Medical Image Analysis*, 102:103551, May 2025.

- [40] Andreas Steiner, Alexander Kolesnikov, Xiaohua Zhai, Ross Wightman, Jakob Uszkoreit, and Lucas Beyer. How to train your vit? data, augmentation, and regularization in vision transformers. *arXiv preprint arXiv:2106.10270*, 2021.
- [41] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9650–9660, 2021.
- [42] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- [43] Li Fei-Fei, Rob Fergus, and Pietro Perona. Learning generative visual models from few training examples: An incremental bayesian approach tested on 101 object categories. *Computer Vision and Image Understanding*, 106(1):59–70, 2007.
- [44] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- [45] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman. The PASCAL Visual Object Classes Challenge 2012 (VOC2012) Results. <http://www.pascal-network.org/challenges/VOC/voc2012/workshop/index.html>.
- [46] Paul Bergmann, Kilian Batzner, Michael Fauser, David Sattlegger, and Carsten Steger. The mvtec anomaly detection dataset: a comprehensive real-world dataset for unsupervised anomaly detection. *International Journal of Computer Vision*, 129(4):1038–1059, 2021.
- [47] Joseph Paul Cohen, Paul Morrison, and Lan Dao. Covid-19 image data collection, 2020.
- [48] Xiran Wang, Jian Zhang, Lei Qi, and Yinghuan Shi. Generalizable decision boundaries: Dualistic meta-learning for open set domain generalization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11564–11573, October 2023.